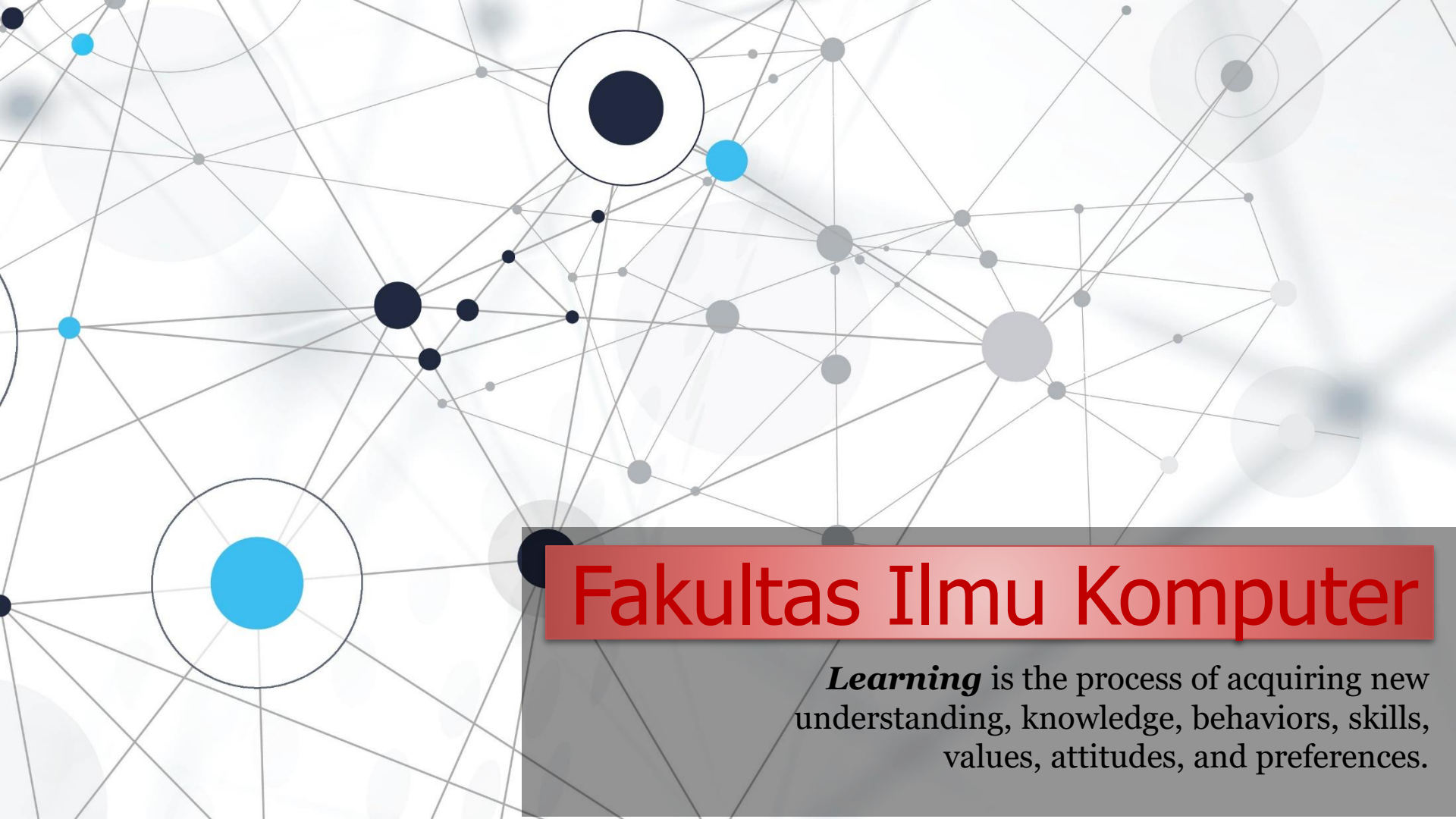


An abstract network diagram with various sized nodes (black, blue, grey) connected by thin grey lines. Some nodes are highlighted with larger circles. The background is white with faint grey circles.

Fakultas Ilmu Komputer

Learning is the process of acquiring new understanding, knowledge, behaviors, skills, values, attitudes, and preferences.

PHP MVC (*model, view, controller*)

Back-end Framework #1 – MVC Architectural Pattern



Learn More about MVC Structure

[Basic MVC File and Directory Structure]



Mengapa keamanan di bagian **back-end website** sangat penting?

```
root@ISC2025: /var/www/htn
104.23.170.85 - [08/Sep/2025:01:20:51 +0000] "GET /.env HTTP/1.1" 404 12910 "-" "Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/81.0.4044.129 Safari/537.36"
104.23.170.85 - [08/Sep/2025:01:20:52 +0000] "GET /.env.save HTTP/1.1" 404 12909 "-" "Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/81.0.4044.129 Safari/537.36"
172.71.103.188 - [08/Sep/2025:01:20:52 +0000] "GET /.env.prod HTTP/1.1" 404 12910 "-" "Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/81.0.4044.129 Safari/537.36"
104.23.170.85 - [08/Sep/2025:01:20:52 +0000] "GET /api/.env HTTP/1.1" 404 12909 "-" "Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/81.0.4044.129 Safari/537.36"
172.71.95.125 - [08/Sep/2025:01:20:53 +0000] "GET /dev/.env HTTP/1.1" 404 12910 "-" "Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/81.0.4044.129 Safari/537.36"
104.23.168.80 - [08/Sep/2025:01:20:54 +0000] "GET /application/.env HTTP/1.1" 404 12910 "-" "Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/81.0.4044.129 Safari/537.36"
104.23.166.29 - [08/Sep/2025:01:20:57 +0000] "GET /backend/.env HTTP/1.1" 404 12910 "-" "Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/81.0.4044.129 Safari/537.36"
172.71.103.187 - [08/Sep/2025:01:20:59 +0000] "GET /.env.example HTTP/1.1" 404 12910 "-" "Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/81.0.4044.129 Safari/537.36"
172.70.46.160 - [08/Sep/2025:01:20:59 +0000] "GET /admin/.env HTTP/1.1" 404 12910 "-" "Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/81.0.4044.129 Safari/537.36"
54.172.164.173 - [08/Sep/2025:01:24:09 +0000] "GET /shell.php HTTP/1.1" 301 539 "-" "-"
54.172.164.173 - [08/Sep/2025:01:24:09 +0000] "GET /shell.php HTTP/1.1" 200 12918 "-" "-"
54.172.164.173 - [08/Sep/2025:01:24:13 +0000] "GET /wp-login.php HTTP/1.1" 301 545 "-" "-"
54.172.164.173 - [08/Sep/2025:01:24:13 +0000] "GET /wp-login.php HTTP/1.1" 200 12918 "-" "-"
54.172.164.173 - [08/Sep/2025:01:24:21 +0000] "GET /wp-admin/includes/class_api.php HTTP/1.1" 301 583 "-" "-"
54.172.164.173 - [08/Sep/2025:01:24:21 +0000] "GET /wp-admin/includes/class_api.php HTTP/1.1" 200 12918 "-" "-"
54.172.164.173 - [08/Sep/2025:01:24:26 +0000] "GET /wp-admin/dropdown.php HTTP/1.1" 301 563 "-" "-"
54.172.164.173 - [08/Sep/2025:01:24:27 +0000] "GET /wp-admin/dropdown.php HTTP/1.1" 200 12918 "-" "-"
54.172.164.173 - [08/Sep/2025:01:24:27 +0000] "GET /wp-admin/css/index.php HTTP/1.1" 301 565 "-" "-"
54.172.164.173 - [08/Sep/2025:01:24:27 +0000] "GET /wp-admin/css/index.php HTTP/1.1" 200 12918 "-" "-"
54.172.164.173 - [08/Sep/2025:01:24:30 +0000] "GET /wp-admin/images/index.php HTTP/1.1" 301 571 "-" "-"
54.172.164.173 - [08/Sep/2025:01:24:30 +0000] "GET /wp-admin/images/index.php HTTP/1.1" 200 12918 "-" "-"
54.172.164.173 - [08/Sep/2025:01:24:30 +0000] "GET /wp-admin/css/colors/index.php HTTP/1.1" 301 579 "-" "-"
54.172.164.173 - [08/Sep/2025:01:24:30 +0000] "GET /wp-admin/css/colors/index.php HTTP/1.1" 200 12918 "-" "-"
54.172.164.173 - [08/Sep/2025:01:24:35 +0000] "GET /wp-admin/network/xmlrpc.php HTTP/1.1" 301 581 "-" "-"
54.172.164.173 - [08/Sep/2025:01:24:35 +0000] "GET /wp-admin/network/xmlrpc.php HTTP/1.1" 200 12918 "-" "-"
54.172.164.173 - [08/Sep/2025:01:24:36 +0000] "GET /wp-admin/user/xmlrpc.php HTTP/1.1" 301 575 "-" "-"
54.172.164.173 - [08/Sep/2025:01:24:36 +0000] "GET /wp-admin/user/xmlrpc.php HTTP/1.1" 200 12918 "-" "-"
54.172.164.173 - [08/Sep/2025:01:24:36 +0000] "GET /wp-admin/images/xmlrpc.php HTTP/1.1" 301 579 "-" "-"
54.172.164.173 - [08/Sep/2025:01:24:36 +0000] "GET /wp-admin/images/xmlrpc.php HTTP/1.1" 200 12918 "-" "-"
54.172.164.173 - [08/Sep/2025:01:24:36 +0000] "GET /wp-admin/ccx/index.php HTTP/1.1" 301 565 "-" "-"
54.172.164.173 - [08/Sep/2025:01:24:38 +0000] "GET /wp-admin/ccx/index.php HTTP/1.1" 200 12918 "-" "-"
54.172.164.173 - [08/Sep/2025:01:24:40 +0000] "GET /wp-admin/x.php?action=768776e296b6f286f26796e2a72607e2972647 HTTP/1.1" 301 589 "-" "-"
54.172.164.173 - [08/Sep/2025:01:24:40 +0000] "GET /wp-admin/css/colors/blue/about.php HTTP/1.1" 301 589 "-" "-"
54.172.164.173 - [08/Sep/2025:01:24:41 +0000] "GET /wp-admin/css/colors/blue/about.php HTTP/1.1" 200 12918 "-" "-"
54.172.164.173 - [08/Sep/2025:01:24:45 +0000] "GET /wp-admin/css/colors/coffee/index.php HTTP/1.1" 301 593 "-" "-"
54.172.164.173 - [08/Sep/2025:01:24:46 +0000] "GET /wp-admin/css/colors/coffee/index.php HTTP/1.1" 200 12918 "-" "-"
54.172.164.173 - [08/Sep/2025:01:24:48 +0000] "GET /wp-admin/js/about.php HTTP/1.1" 301 563 "-" "-"
54.172.164.173 - [08/Sep/2025:01:24:48 +0000] "GET /wp-admin/js/about.php HTTP/1.1" 200 12918 "-" "-"
54.172.164.173 - [08/Sep/2025:01:24:54 +0000] "GET /wp-admin/network/cloud.php HTTP/1.1" 301 573 "-" "-"
54.172.164.173 - [08/Sep/2025:01:24:54 +0000] "GET /wp-admin/network/cloud.php HTTP/1.1" 200 12918 "-" "-"
54.172.164.173 - [08/Sep/2025:01:24:55 +0000] "GET /wp-admin/images/about.php HTTP/1.1" 301 571 "-" "-"
54.172.164.173 - [08/Sep/2025:01:24:55 +0000] "GET /wp-admin/images/about.php HTTP/1.1" 200 12918 "-" "-"
54.172.164.173 - [08/Sep/2025:01:24:55 +0000] "GET /wp-admin/css/about.php HTTP/1.1" 301 565 "-" "-"
54.172.164.173 - [08/Sep/2025:01:24:56 +0000] "GET /wp-admin/css/about.php HTTP/1.1" 200 12918 "-" "-"
54.172.164.173 - [08/Sep/2025:01:25:02 +0000] "GET /wp-admin/js/about.php HTTP/1.1" 301 563 "-" "-"
54.172.164.173 - [08/Sep/2025:01:25:02 +0000] "GET /wp-admin/js/about.php HTTP/1.1" 200 12918 "-" "-"
54.172.164.173 - [08/Sep/2025:01:25:04 +0000] "GET /wp-admin/includes/about.php HTTP/1.1" 301 575 "-" "-"
54.172.164.173 - [08/Sep/2025:01:25:05 +0000] "GET /wp-admin/includes/about.php HTTP/1.1" 200 12918 "-" "-"
54.172.164.173 - [08/Sep/2025:01:25:11 +0000] "GET /shell20211028.php HTTP/1.1" 301 555 "-" "-"
54.172.164.173 - [08/Sep/2025:01:25:11 +0000] "GET /shell20211028.php HTTP/1.1" 200 12918 "-" "-"
54.172.164.173 - [08/Sep/2025:01:25:12 +0000] "GET /wp-content/shell20211028.php HTTP/1.1" 301 577 "-" "-"
```

Amankan .env

```
<Files ".env*">
```

Require all denied

```
</Files>
```

Ini upaya attacker untuk membaca file konfigurasi **.env** (biasa dipakai di Laravel, Symfony, Node.js, dsb) dari sisi front-end. Directory **.env** sering berisi **database password, API key, SMTP credential**.

Untungnya, semua request hasilnya 404 → artinya file tidak ada (aman).

Kalau file **.env** memang ada dan tidak dilindungi, maka data sensitif bisa bocor.

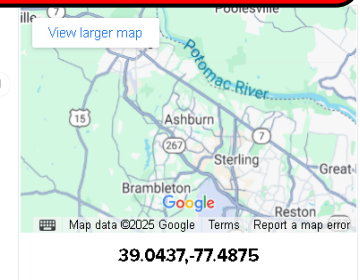
Akses ke .env :

- ✓ GET /.env
- ✓ GET /.env.save
- ✓ GET /.env.prod
- ✓ GET /api/.env
- ✓ GET /dev/.env
- ✓ GET /application/.env
- ✓ GET /backend/.env
- ✓ GET /.env.example
- ✓ GET /admin/.env

Attacker mencoba akses file-file PHP di dalam folder WordPress (/wp-admin/, /wp-content/). Dimana, tujuannya adalah untuk menemukan plugin/theme/backdoor yang rentan.

IP Geolocation

City	Ashburn
State	Virginia
Country	United States
Postal	20147
Local time	11:22 PM, Sunday, September 07, 2025
Timezone	America/New_York
Coordinates	39.0437,-77.4875



Remember, attacks from the back-end side can also occur. These attacks target the server or application layer behind the scenes, not just the web's front-end. In other words, what is being attacked is the server system, database, API, or application logic.



```
Sep 07 21:19:13 ISC2025 sshd[18165]: User root from 31.182.126.68 not allowed because not listed in AllowUsers
Sep 07 21:19:13 ISC2025 sshd[18165]: Connection closed by invalid user root 31.182.126.68 port 34746 [preauth]
Sep 07 21:19:28 ISC2025 sshd[18167]: Invalid user deepface1 from 31.182.126.68 port 38762
Sep 07 21:19:28 ISC2025 sshd[18167]: Connection closed by invalid user deepface1 31.182.126.68 port 38762 [preauth]
Sep 07 21:19:44 ISC2025 sshd[18172]: User root from 31.182.126.68 not allowed because not listed in AllowUsers
Sep 07 21:19:44 ISC2025 sshd[18172]: Connection closed by invalid user root 31.182.126.68 port 55606 [preauth]
^C
root@ISC2025:/home/ISC2025# sudo fail2ban-client status sshd
Status for the jail: sshd
|- Filter
| |- Currently failed: 1
| |- Total failed: 1653
| '- File list: /var/log/auth.log
'- Actions
  |- Currently banned: 8
  |- Total banned: 9
  '- Banned IP list: 89.0.111.227 222.241.248.136 61.177.238.130 2_54.47.72 14.18.190.138 171.222.186.114 31.182.126.68 116.22.170.169
```



- 1) By **strengthening back-end security**, we can reduce the risk of attacks such as SQL injection, cross-site scripting (XSS), and attacks against software flaws (*kelemahan software*).
- 2) The back-end is also responsible for **managing servers, databases**, and other **infrastructure** used in website operations. **Strong security** is required to protect servers from DDoS attacks, theft of server resources, and other threats to the infrastructure.

The purpose of the "**hardening**" process on an Ubuntu 22.04 server is to **improve system security** by **reducing potential security risks** and closing loopholes that can be exploited by attackers. This includes measures to lock down the system, mitigate potential attacks, and minimize opportunities for unauthorized access.

- 1) Website **FRAMEWORK** with more security config approach.
- 2) **MySQL root** sudah dibuat password yg complex (sudo mysql_secure_installation).
- 3) MySQL user tanpa "**default user name**".
- 4) **Firewall** status activated (**80, 443 allowed**). Menggunakan UFW (Uncomplicated Firewall) atau iptables untuk mengkonfigurasi aturan firewall yang sesuai untuk server, dan hanya membuka port yang diperlukan.
- 5) SSH *port* sudah tidak menggunakan yang default (**port 22**).
- 6) **User root password** changed.
- 7) **SSH root access** disabled (**PermitRootLogin no**).
- 8) Sudah menggunakan **public dan private RSA key** untuk otentikasi SSH. Mencegah serangan **BRUTE FORCE** pada layanan yang memerlukan otentikasi, seperti SSH, dengan mengimplementasikan mekanisme pembatasan login dan penggunaan kunci SSH.
- 9) Http to **HTTPS connection** (SSL certificate).

Apakah “hardening” seperti atas sudah cukup di Ubuntu 22.04 server?

The slide features a white background with green leaves and branches framing the top and bottom edges. The leaves are vibrant green and appear to be from a tree or shrub. The text is centered in the middle of the slide.

About MVC Framework

Model-View-Controller (MVC)

The Model-View-Controller (MVC) framework is an **architectural pattern** that separates an application into three main logical components Model, View, and Controller. Each architecture component is built to handle specific development aspect of an application.

Most used architectural pattern for today's web applications

Three Important MVC Components (*layers*)

Component #1 → Model :

- ✓ The model component stores data and its related logic. It represents data that is being transferred between controller components or any other related business logic. For example, a Controller object will retrieve the customer info from the database. It manipulates data and send back to the database or use it to render the same data.
- ✓ It responds to the request from the views and also responds to instructions from the controller to update itself. It is also the lowest level of the pattern which is responsible for maintaining data.

Component #2 → View :

- ✓ The View is that part of the application that represents the presentation of data.
- ✓ Views are created by the data collected from the model data. A view requests the model to give information so that it resents the output presentation to the user.
- ✓ The view also represents the data from chats, diagrams, and table. For example, any customer view will include all the UI components like text boxes, drop downs, etc.

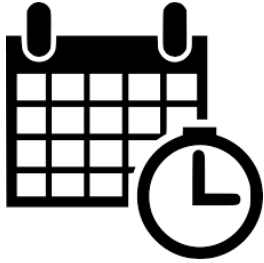
Component #3 → Controller :

- ✓ The Controller is that part of the application that handles the user interaction. The controller interprets the mouse and keyboard inputs from the user, informing model and the view to change as appropriate.
- ✓ A Controller send's commands to the model to update its state (*E.g., Saving a specific document*). The controller also sends commands to its associated view to change the view's presentation (*For example scrolling a particular document*).

The paper was published under the title:

Burbeck, Steve. "Applications programming in smalltalk-80 (tm): How to use model-view-controller (mvc)." *Smalltalk-80 v2 5* (1992): 1-11.

Three Important MVC Components (*layers*)



Model: It includes all the data and its related logic.



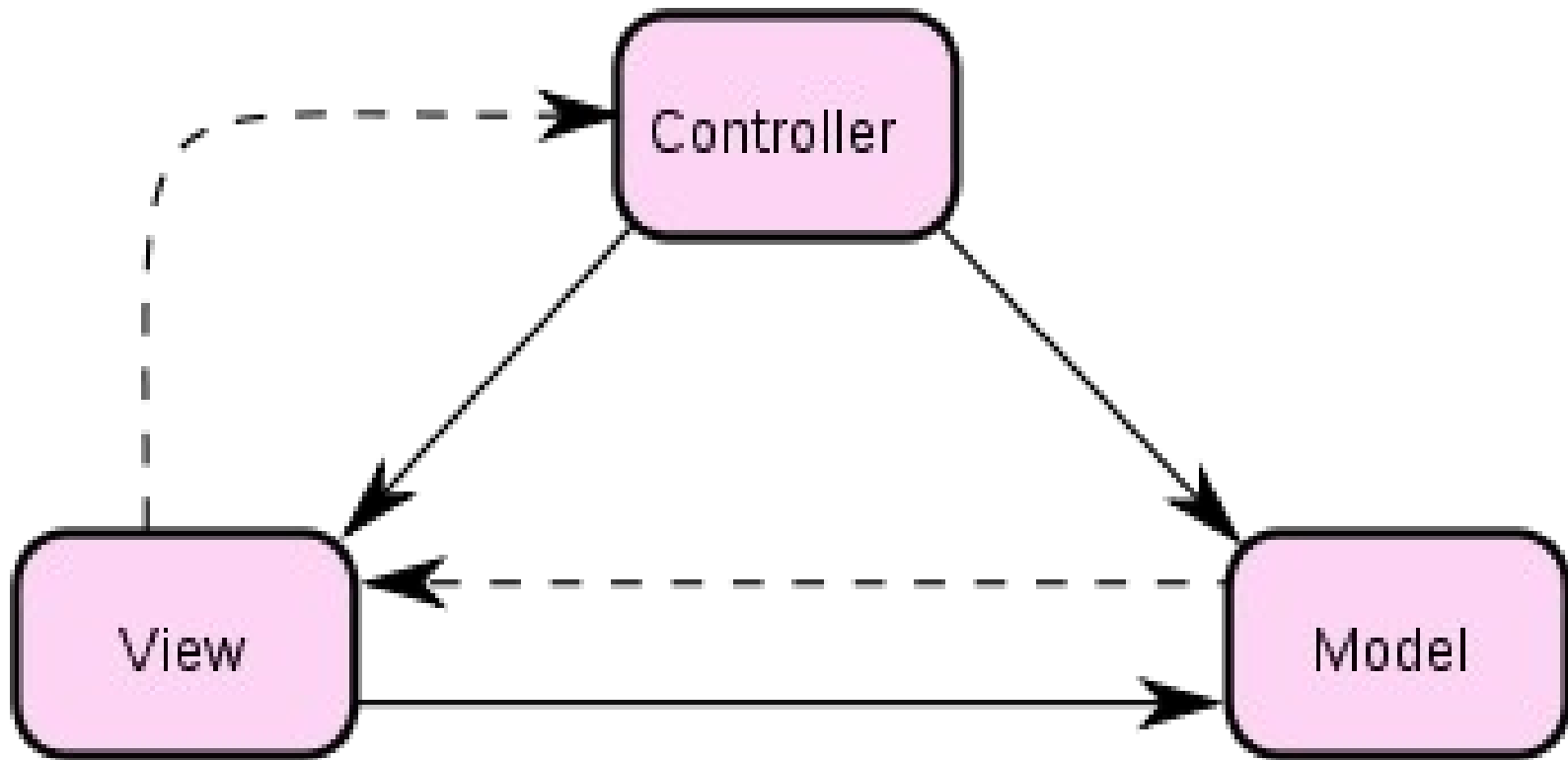
View: Present data to the user or handles user interaction.



Controller: An interface between Model and View components.

Three Important MVC Components (*layers*)

Logic program (*controller*)



Tampilan (*view*) => UI

Manage data

Component #1 $\Rightarrow$ Model



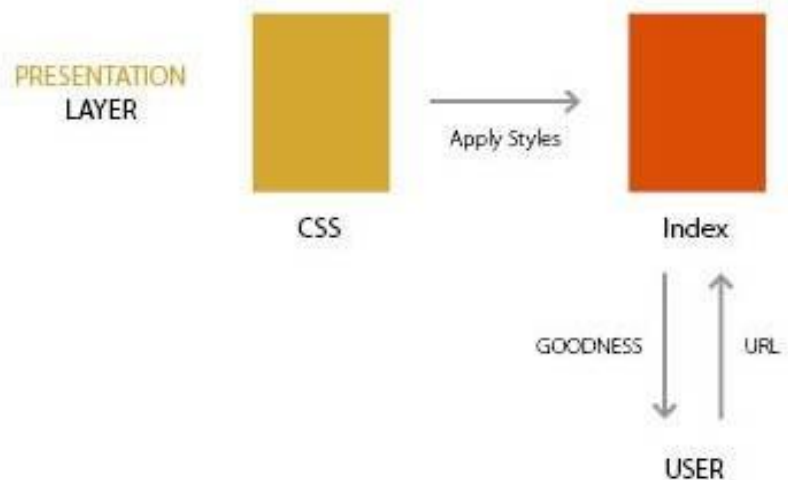
- ❑ Responsible for managing the data.
- ❑ Stores and retrieves entities used by an application, usually from a **database** but can also include invocations (*call/import*) of external web services or APIs.
- ❑ Contains the logic implemented by the application (*business logic*)

Component #2 $\Rightarrow$ View (Presentation)



Responsible to display the data provided by the model in a specific format.

Presentation Layer



Component #3 $\Rightarrow$ Controller



- ❑ Handles the model and view layers to work together.
- ❑ Receives a request from the client, invokes the model to perform the requested operations and sends the data to the view.



Menggunakan Fungsi **`require_once()`**

Perbedaan Fungsi *include* dan *require*

- ❑ **Fungsi `include()`** akan menyertakan dan mengevaluasi seluruh program yang ada di *file* yang disertakan. Jika terdapat *error* pada program yang disertakan, maka *error* akan ditampilkan di layar.
 - ✓ Jika *file* yang disertakan ternyata tidak ditemukan (*mungkin karena lokasi yang salah atau memang file tidak ada*), maka program selanjutnya (*setelah include*) akan tetap dijalankan walaupun ditampilkan *error*.
- ❑ **Fungsi `require()`** pada dasarnya sama dengan perintah `include()`. Perbedaan hanya terletak pada saat *file* yang disertakan tidak ditemukan, maka perintah-perintah selanjutnya tidak akan dijalankan.

Perbedaan Antara `Include_once` dan `Require_once`

Penambahan akhiran **`_once`** baik pada *include* maupun *require* akan memastikan bahwa ***file*** yang disertakan **hanya dieksekusi sekali**, walaupun file disertakan beberapa kali. Sebaliknya tanpa akhiran **`_once`** penyertaan ***file*** yang sama beberapa kali mungkin akan menyebabkan ***error***.

Kesalahan Penggunaan *include* dan *require*

Langkah #1 : Buat file *Schools.class.php*

```
C:\xampp\htdocs\lecture9\latihan1\Schools.class.php - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ? X
Schools.class.php
1 <?php
2 class Schools {
3     var $student_id = 1000;
4     function getStudent() {
5         return $student_id;
6     }
7 }
8 ?>
```

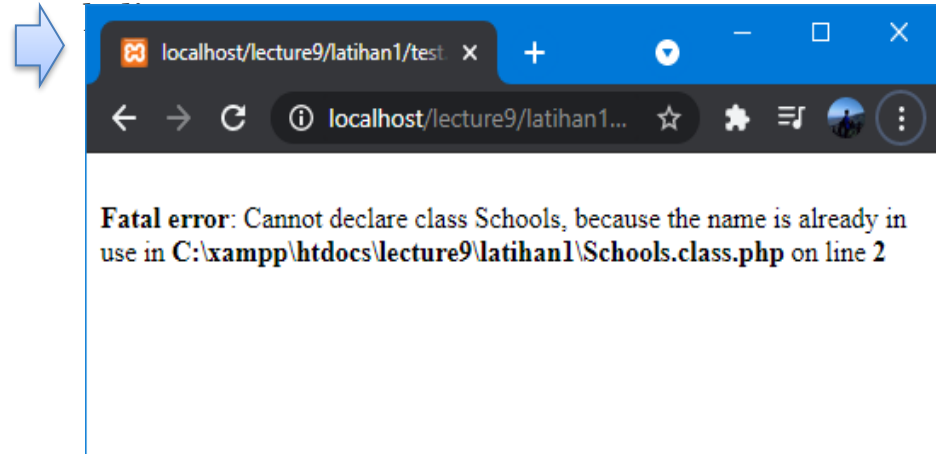
Langkah #2 : Buat file *test.php*

```
C:\xampp\htdocs\lecture9\latihan1\test.php - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ? X
test.php
1 <?php
2 require "Schools.class.php";
3 require "Schools.class.php";
4 echo "Program setelah pemanggilan.";
5 ?>
```

Langkah #3 : Modify/Change file *test.php*

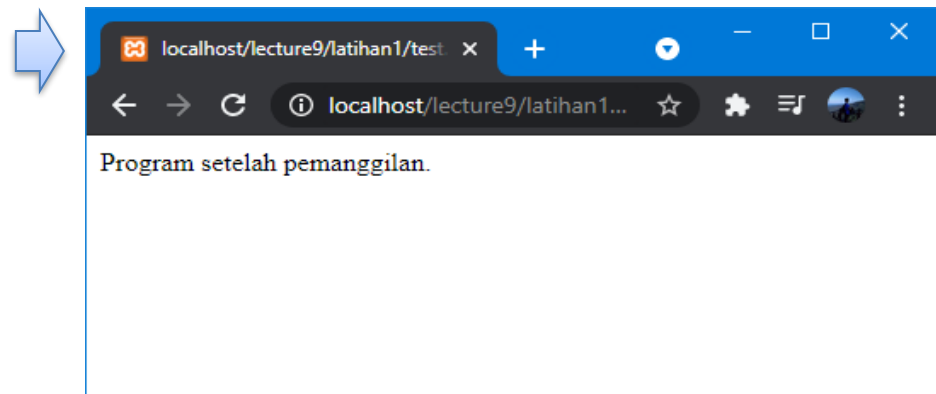
```
C:\xampp\htdocs\lecture9\latihan1\test.php - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ? X
test.php
1 <?php
2 require_once "Schools.class.php";
3 require_once "Schools.class.php";
4 echo "Program setelah pemanggilan.";
5 ?>
```

Output #1 : Error class Schools di *declare* dua



Untuk menghindari *error*, tambahkan akhiran *once* di fungsi *include* maupun *require* sehingga program hanya akan mengeksekusi penyertaan *file* sebanyak sekali. Ganti fungsi *require* pada baris 2 dan 3 program *test.php* di samping dengan fungsi *require_once*.

Output #2 : Setelah menambahkan akhiran *once*





Follow the steps on the next slide!!!

A decorative border of green leaves and branches at the top of the page.

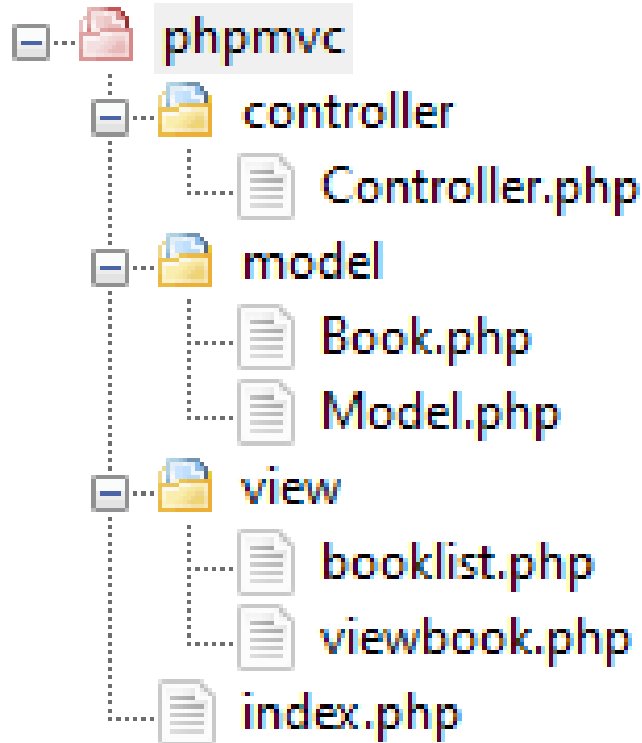
Simple MVC Application

*To code our own MVC framework from scratch
(from the very beginning without utilizing or relying on any previous framework)*

A decorative border of green leaves and branches at the bottom of the page.

Sample MVC Application

Basic File and Directory Structures:



Terdapat dua tujuan dari pengembangan framework yaitu ***pengorganisasian file*** dan ***templating***.

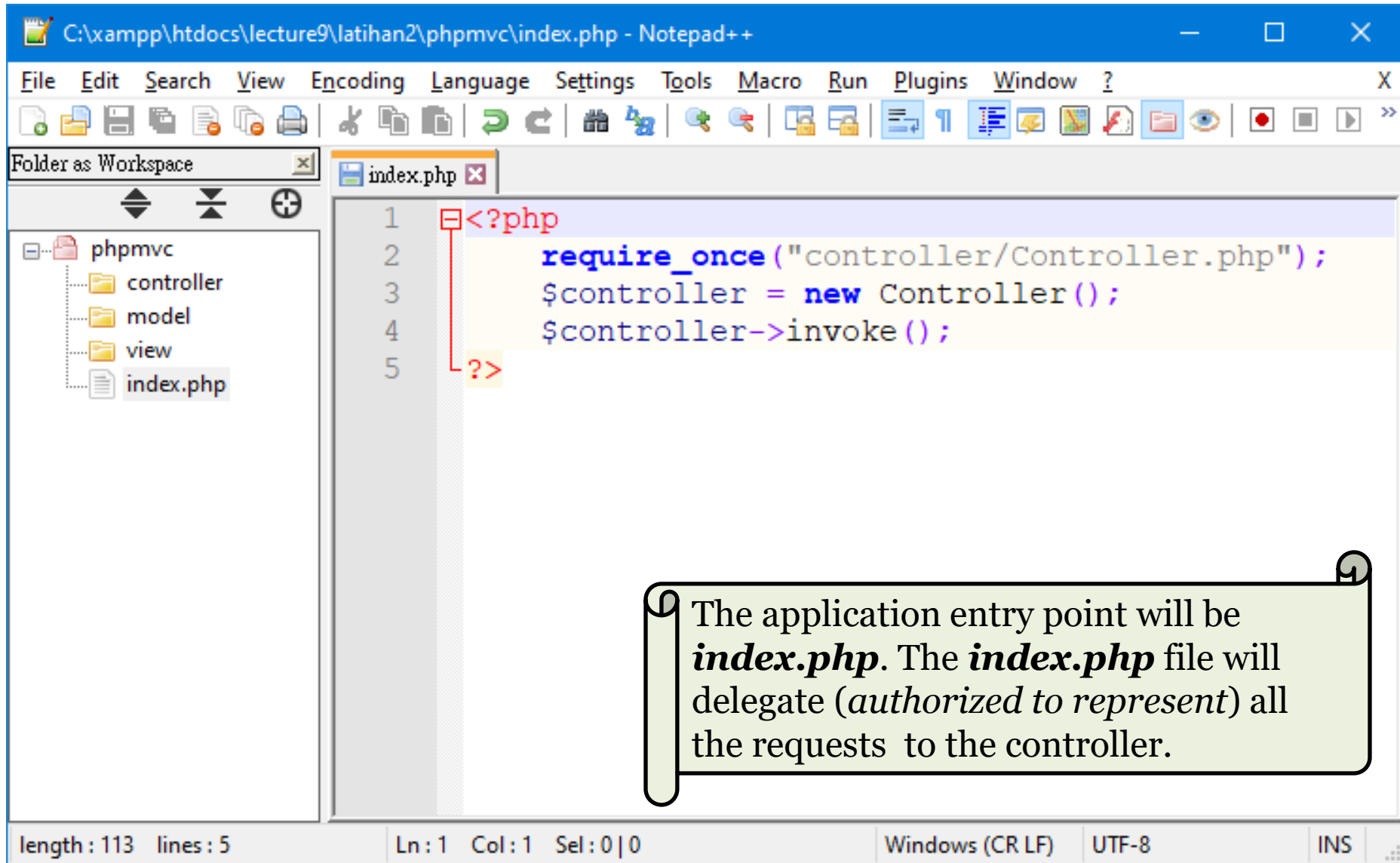
- (1) **Pengorganisasian File**, bertujuan untuk memisahkan antara logic program (*controller*) dengan tampilannya (*view*) dan agar aplikasi menjadi lebih rapi dan mudah dibaca.
- (2) **Templating**, seringkali aplikasi memiliki outline (*layout*) yang sama untuk setiap page, yang berubah hanya kontennya saja. Oleh karena itu kita dapat menerapkan *layouting* pada page di *website*.

The Model and View are separated, making the application more flexible.



Create *index.php* File

Entry point - ***index.php***



The screenshot shows a Notepad++ window with the file `C:\xampp\htdocs\lecture9\latihan2\phpmvc\index.php` open. The left sidebar shows a folder structure with `phpmvc` containing `controller`, `model`, `view`, and `index.php`. The main editor displays the following PHP code:

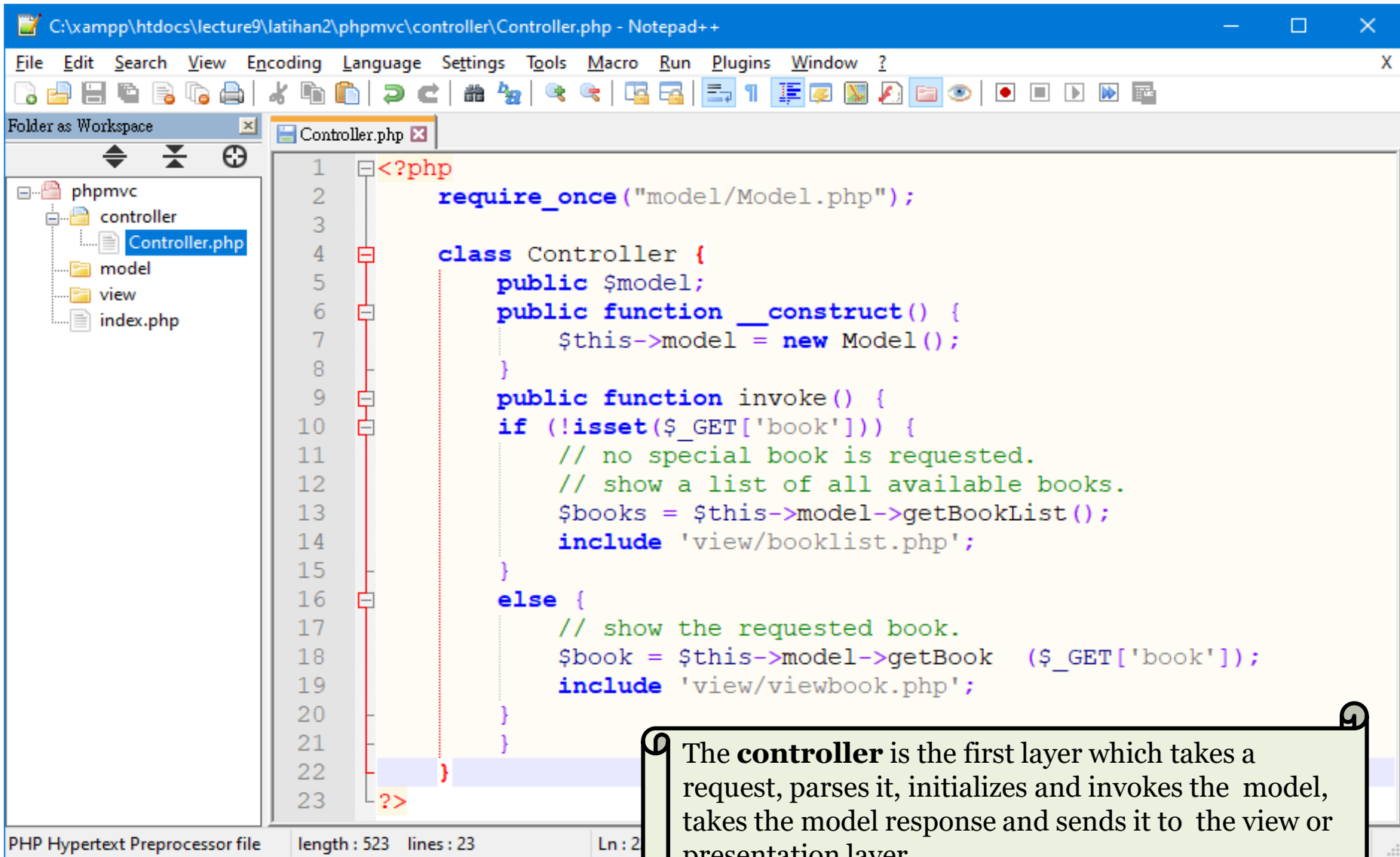
```
1 <?php
2     require_once("controller/Controller.php");
3     $controller = new Controller();
4     $controller->invoke();
5 ?>
```

A red bracket on the left side of the code block indicates the entire content. A callout box in the bottom right corner contains the following text:

The application entry point will be ***index.php***. The ***index.php*** file will delegate (*authorized to represent*) all the requests to the controller.

length : 113 lines : 5
Ln : 1 Col : 1 Sel : 0 | 0
Windows (CR LF) UTF-8 INS

Controller – controller/Controller.php



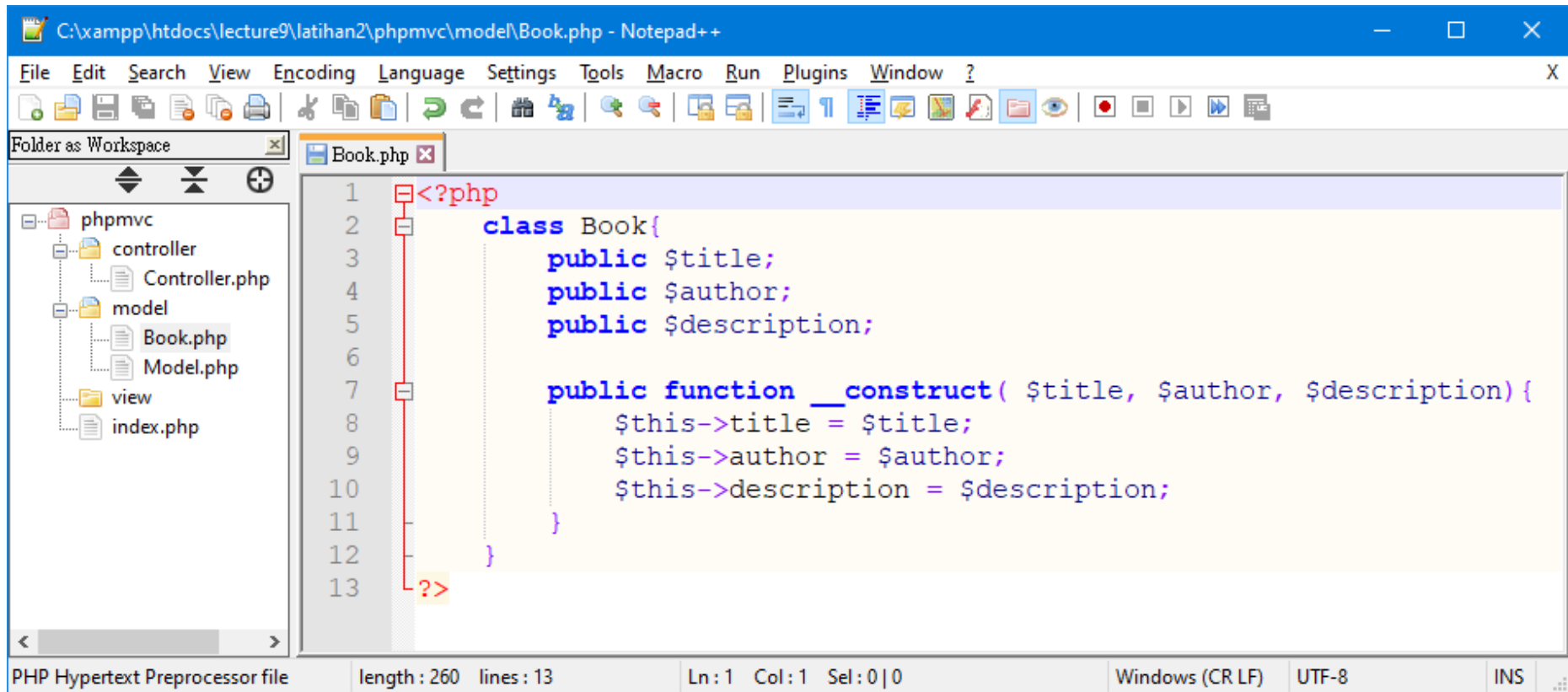
The screenshot shows a Notepad++ window titled "C:\xampp\htdocs\lecture9\latihan2\phpmvc\controller\Controller.php - Notepad++". The editor displays the following PHP code:

```
1 <?php
2     require_once("model/Model.php");
3
4     class Controller {
5     public $model;
6     public function __construct() {
7         $this->model = new Model();
8     }
9     public function invoke() {
10        if (!isset($_GET['book'])) {
11            // no special book is requested.
12            // show a list of all available books.
13            $books = $this->model->getBookList();
14            include 'view/booklist.php';
15        }
16        else {
17            // show the requested book.
18            $book = $this->model->getBook($_GET['book']);
19            include 'view/viewbook.php';
20        }
21    }
22 }
23 ?>
```

The left sidebar shows the project structure: phpmvc > controller > Controller.php. The status bar at the bottom indicates "PHP Hypertext Preprocessor file", "length : 523", "lines : 23", and "Ln : 2".

The **controller** is the first layer which takes a request, parses it, initializes and invokes the model, takes the model response and sends it to the view or presentation layer.

Model – model/Book.php



The screenshot shows a Notepad++ window titled "C:\xampp\htdocs\lecture9\latihan2\phpmvc\model\Book.php - Notepad++". The left sidebar displays a "Folder as Workspace" view of the project structure:

- phpmvc
 - controller
 - Controller.php
 - model
 - Book.php
 - Model.php
 - view
 - index.php

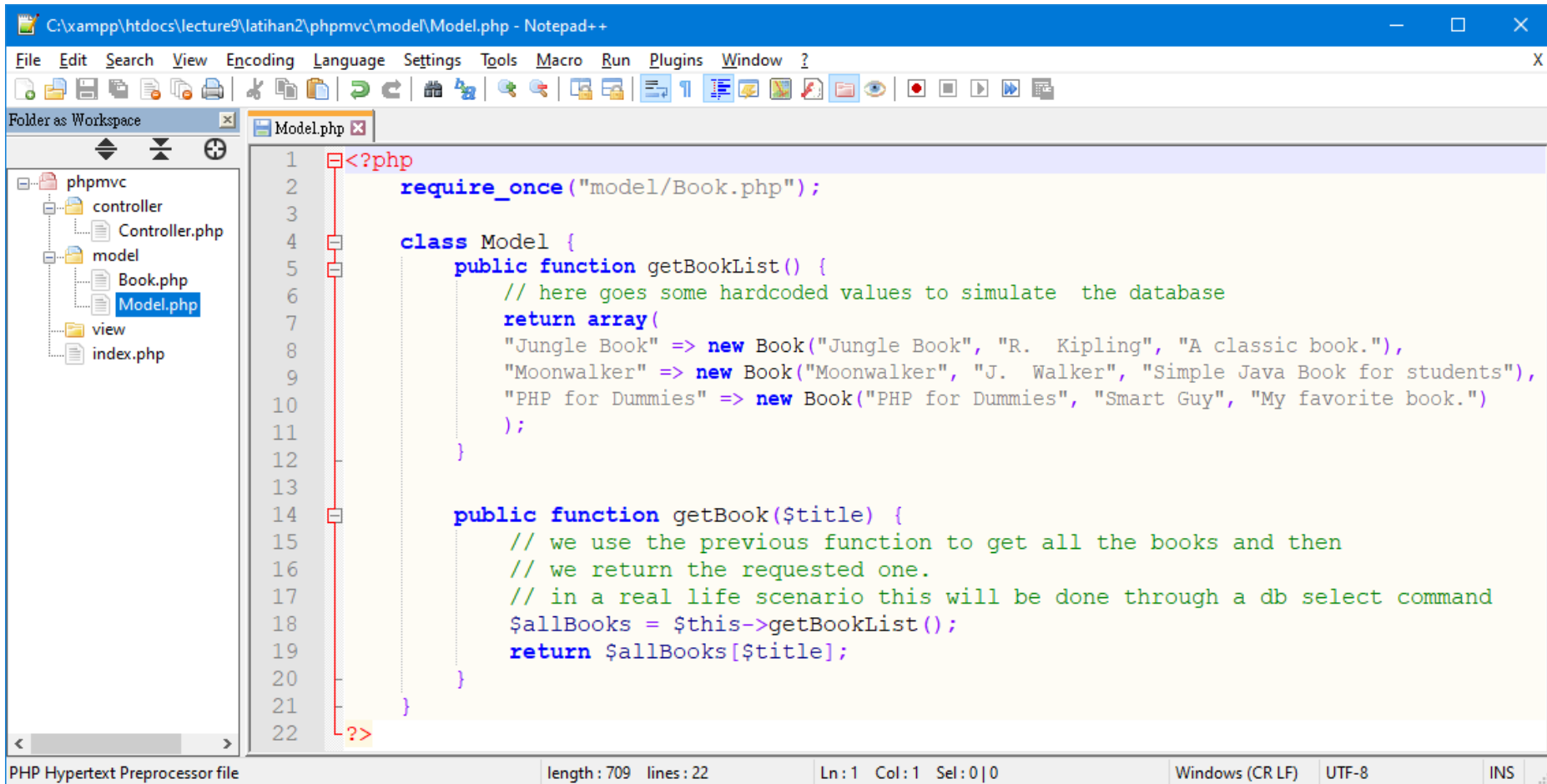
The main editor area shows the code for Book.php:

```
1 <?php
2 class Book{
3     public $title;
4     public $author;
5     public $description;
6
7     public function __construct( $title, $author, $description){
8         $this->title = $title;
9         $this->author = $author;
10        $this->description = $description;
11    }
12 }
13 ?>
```

The status bar at the bottom indicates: "PHP Hypertext Preprocessor file", "length : 260 lines : 13", "Ln : 1 Col : 1 Sel : 0 | 0", "Windows (CR LF)", "UTF-8", and "INS".

The purpose is to keep data. Depending on the implementation, entity objects can be replaced by XML or JSON chunk of data. It is recommended that entities do not **encapsulate any business logic**.

Model – model/Model.php

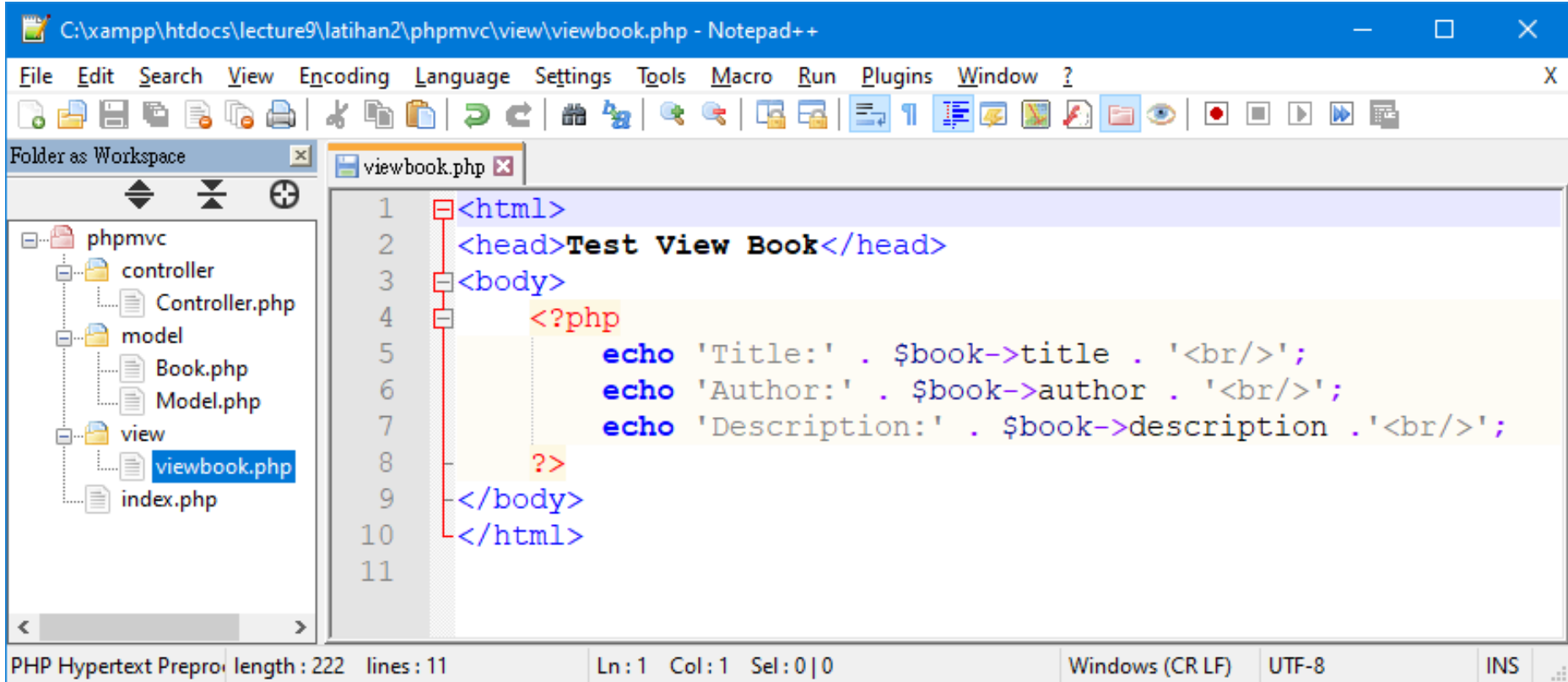


```
1 <?php
2     require_once("model/Book.php");
3
4     class Model {
5     public function getBookList() {
6         // here goes some hardcoded values to simulate the database
7         return array(
8             "Jungle Book" => new Book("Jungle Book", "R. Kipling", "A classic book."),
9             "Moonwalker" => new Book("Moonwalker", "J. Walker", "Simple Java Book for students"),
10            "PHP for Dummies" => new Book("PHP for Dummies", "Smart Guy", "My favorite book.")
11        );
12    }
13
14    public function getBook($title) {
15        // we use the previous function to get all the books and then
16        // we return the requested one.
17        // in a real life scenario this will be done through a db select command
18        $allBooks = $this->getBookList();
19        return $allBooks[$title];
20    }
21 }
22 ?>
```

PHP Hypertext Preprocessor file length : 709 lines : 22 Ln : 1 Col : 1 Sel : 0 | 0 Windows (CR LF) UTF-8 INS

In a real-world scenario, the model will include all the entities and the classes to persist data into the database, and the classes encapsulating the *business logic*.

View – view/viewbook.php



The screenshot shows the Notepad++ editor with the file `C:\xampp\htdocs\lecture9\latihan2\phpmvc\view\viewbook.php` open. The editor displays the following code:

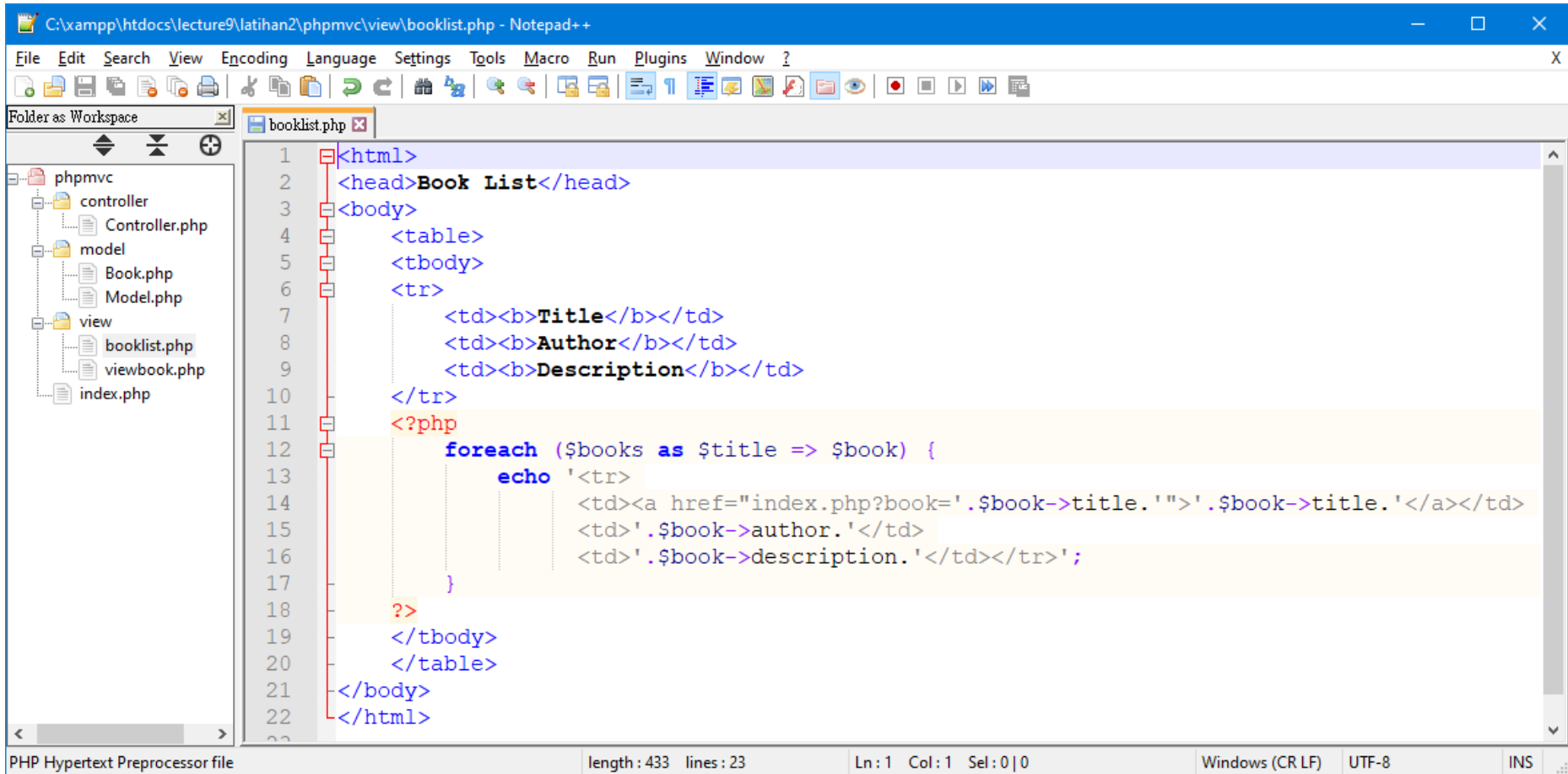
```
1 <html>
2 <head>Test View Book</head>
3 <body>
4 <?php
5     echo 'Title:' . $book->title . '<br/>';
6     echo 'Author:' . $book->author . '<br/>';
7     echo 'Description:' . $book->description . '<br/>';
8 ?>
9 </body>
10 </html>
11
```

The left sidebar shows the project structure:

- phpmvc
 - controller
 - Controller.php
 - model
 - Book.php
 - Model.php
 - view
 - viewbook.php (selected)
 - index.php

The status bar at the bottom indicates: PHP Hypertext Preprocessor, length : 222, lines : 11, Ln : 1 Col : 1 Sel : 0 | 0, Windows (CR LF), UTF-8, INS.

View – view/booklist.php



```
1 <html>
2 <head>Book List</head>
3 <body>
4 <table>
5 <tbody>
6 <tr>
7 <td><b>Title</b></td>
8 <td><b>Author</b></td>
9 <td><b>Description</b></td>
10 </tr>
11 <?php
12 <foreach ($books as $title => $book) {
13 <echo ' <tr>
14 <td><a href="index.php?book='.$book->title.'">'.$book->title.'</a></td>
15 <td>'.$book->author.'</td>
16 <td>'.$book->description.'</td></tr>';
17 }
18 ?>
19 </tbody>
20 </table>
21 </body>
22 </html>
```

Folder as Workspace

phpmvc

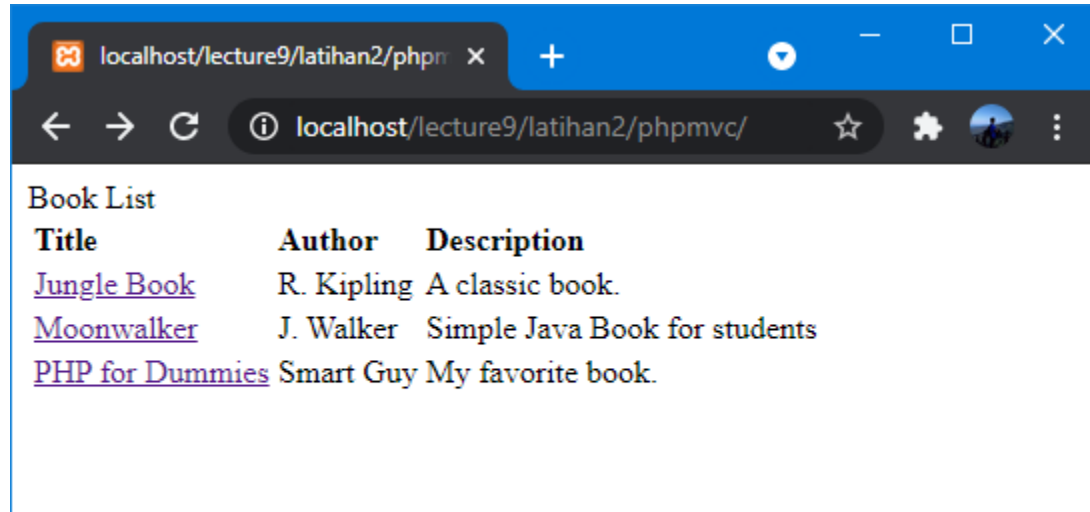
- controller
 - Controller.php
- model
 - Book.php
 - Model.php
- view
 - booklist.php
 - viewbook.php
 - index.php

PHP Hypertext Preprocessor file | length : 433 | lines : 23 | Ln : 1 | Col : 1 | Sel : 0 | 0 | Windows (CR LF) | UTF-8 | INS

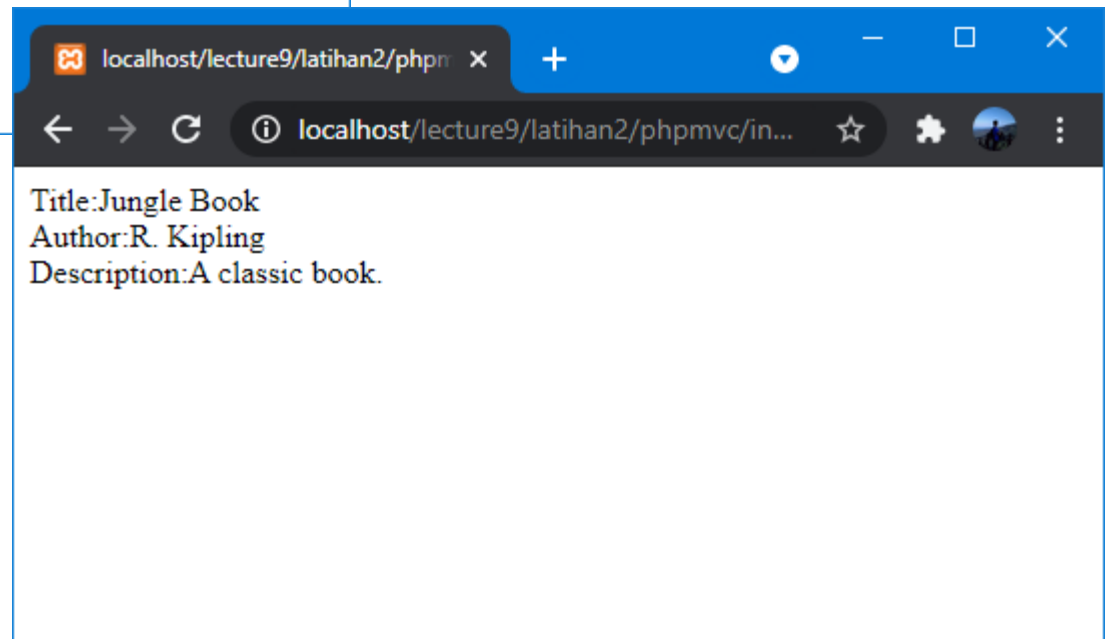
Responsible for formatting the data received from the model in a form accessible to the user. The data can come in different formats from the model or *simple objects* (sometimes called *Value Objects*), XML structures, JSON, etc.

Final Output ⇒ Try your self

Link: <http://localhost/lecture9/latihan2/phpmvc/>



**After clicked one
item in the list.**





Exercise *for* Students

Latihan untuk *Students*

Create directory : C:\xampp\htdocs\lecture9\latihan_{1,2}\
Create file : `_.php`

Ikuti langkah-langkahnya dan buat contoh penggunaan fungsi **`require_once()`** dan juga "**Aplikasi MVC**" **sederhana** yang telah kita pelajari sebelumnya. Tambahkan juga data buku di bawah ini.

Follow the steps and create an example of using the `require_once()` function and also simple "MVC Application" that we've learned earlier. Also add some of the book data below.

Tambahkan 10 data buku di bawah ini:

Title Book	Author	Description
The Great Gatsby	F. Scott Fitzgerald	Classic novel set in the Roaring Twenties
To Kill a Mockingbird	Harper Lee	Pulitzer Prize-winning novel addressing racial injustice
Totalitarianism	George Orwell	Dystopian fiction exploring totalitarianism
Pride and Prejudice	Jane Austen	Romantic novel with social commentary
The Catcher in the Rye	J.D. Salinger	Coming-of-age story of a teenager in New York City
The Hobbit	J.R.R. Tolkien	Fantasy adventure novel featuring Bilbo Baggins
The Da Vinci Code	Dan Brown	Mystery thriller involving a symbologist
Harry Potter series	J.K. Rowling	Fantasy series following the life of Harry Potter
The Alchemist	Paulo Coelho	Philosophical novel about personal legend and destiny
The Hunger Games	Suzanne Collins	Dystopian trilogy set in a post-apocalyptic world

Reference (#1)

MVC Framework Tutorial for Beginners

<https://www.guru99.com/mvc-tutorial.html>

Write your own PHP MVC Framework

<http://anantgarg.com/2009/03/13/write-your-own-php-mvc-framework-part-1/>

Memahami Include, Include_once, Require, dan Require_once Pada PHP

https://jagowebdev.com/include-include_once-require-dan-require_once-pada-php/

PHP: Perbedaan Fungsi include(), require(), include_once() dan require_once()

https://achmatim.net/2013/10/20/php-perbedaan-fungsi-include-require-include_once-dan-require_once/

Perbedaan Include, Include_once, Require Dan Require_once

https://gopursodik.wordpress.com/2018/04/12/perbedaan-include-include_once-require-dan-require_once/

Code Your Own PHP MVC Framework

<https://www.codeproject.com/Articles/1080626/Code-Your-Own-PHP-MVC-Framework-in-Hour>

How to get list of subdirectories in PHP

<https://lucidar.me/en/web-dev/how-to-get-subdirectories-in-php/>

Create your own MVC framework in PHP

<https://medium.com/@noufel.gouirhate/create-your-own-mvc-framework-in-php-af7bd1foca19>

MVC Folder/File Structure

<https://www.dreamincode.net/forums/topic/243434-mvc-folderfile-structure/>

END PRESENTATION

Thank you for your attention

Instructor: S – W – T

THANK YOU

